Proofs

- Bernoulli's inequality: For $x > -1$ and integer $n \geq 1$ and with strict inequality when $n \geq 2$ and $x \neq 0$.

$$(1+x)^n \geq 1 + nx$$

Proof (induction on n).
Base case $n = 1$: $(1+x)^1 = 1 + x$, so equality holds.
Inductive step: assume $(1+x)^n \geq 1 + nx$. Since $1 + x > 0$, multiplying both sides by $1 + x$ preserves the inequality:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x$$

This completes the induction.
Strictness: let $x \neq 0$. For $n = 2$, $(1+x)^2 = 1 + 2x + x^2 > 1 + 2x$. Now suppose $(1+x)^n > 1 + nx$ for some $n \geq 2$. Multiplying by $1 + x > 0$ gives

$$(1+x)^{n+1} > (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x$$

Hence the inequality is strict for all $n \geq 2$.

- Lagrange's identity: For real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$,

$$\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) - \left(\sum_{i=1}^{n} a_i b_i\right)^2 = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}(a_i b_j - a_j b_i)^2$$

Proof. Expand the right-hand side:

$$\frac{1}{2}\sum_{i,j}(a_i b_j - a_j b_i)^2 = \frac{1}{2}\sum_{i,j} a_i^2 b_j^2 - \sum_{i,j} a_i b_j a_j b_i + \frac{1}{2}\sum_{i,j} a_j^2 b_i^2$$

Swapping the names of the indices i and j in the last sum shows that it equals the first sum, so together they give

$$\sum_{i,j} a_i^2 b_j^2 = \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{j=1}^{n} b_j^2\right)$$

The middle sum factorises as

$$\sum_{i,j}(a_i b_i)(a_j b_j) = \left(\sum_{i=1}^{n} a_i b_i\right)\left(\sum_{j=1}^{n} a_j b_j\right) = \left(\sum_{i=1}^{n} a_i b_i\right)^2$$

Therefore the right-hand side equals

$$\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) - \left(\sum_{i=1}^{n} a_i b_i\right)^2$$

which is the left-hand side.

- Cauchy–Schwarz inequality: For real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$, with equality if and only if $a_i b_j = a_j b_i$ for all i, j (i.e. the vectors are proportional).

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)$$

Proof. By Lagrange's identity,

$$\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) - \left(\sum_{i=1}^{n} a_i b_i\right)^2 = \frac{1}{2}\sum_{i,j}(a_i b_j - a_j b_i)^2$$

Each term $(a_ib_j - a_jb_i)^2$ is the square of a real number, hence nonnegative, so the right-hand side is ≥ 0. This gives the inequality.
Equality holds if and only if the right-hand side is 0, which happens if and only if every term vanishes, i.e. $a_ib_j = a_jb_i$ for all i, j.

- Absorption identity: For integers $n \geq k \geq 1$,

$$k\binom{n}{k} = n\binom{n-1}{k-1}$$

Proof (algebraic).

$$k\binom{n}{k} = k \cdot \frac{n!}{k!\,(n-k)!} = \frac{n!}{(k-1)!\,(n-k)!} = n \cdot \frac{(n-1)!}{(k-1)!\,\big((n-1)-(k-1)\big)!} = n\binom{n-1}{k-1}$$

Proof (combinatorial). Count the pairs (S, c), where S is a k-element subset of $\{1, \ldots, n\}$ and $c \in S$.
Choosing S first and then $c \in S$ gives $\binom{n}{k} \cdot k$ pairs.
Choosing c first (n ways) and then the other $k-1$ elements of S from the remaining $n-1$ gives $n\binom{n-1}{k-1}$ pairs.
Both counts are of the same set, so they are equal.

- Alternating binomial sum: For integers $n \geq 1$,

$$\sum_{i=0}^{n} (-1)^i \binom{n}{i} = 0$$

Proof. By the binomial theorem, $(1+x)^n = \sum_{i=0}^{n} \binom{n}{i} x^i$ for every real x. Setting $x = -1$ gives

$$0 = (1-1)^n = \sum_{i=0}^{n} \binom{n}{i} (-1)^i$$

where $0^n = 0$ because $n \geq 1$.

- Second absorption identity: For integers $n \geq k \geq 2$,

$$k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$$

Proof (from absorption). Apply the absorption identity $k\binom{n}{k} = n\binom{n-1}{k-1}$ to get

$$k(k-1)\binom{n}{k} = (k-1) \cdot n\binom{n-1}{k-1}$$

Apply it again to $\binom{n-1}{k-1}$, with $n-1$ in place of n and $k-1$ in place of k. This is valid since $n-1 \geq k-1 \geq 1$:

$$(k-1)\binom{n-1}{k-1} = (n-1)\binom{n-2}{k-2}$$

Therefore

$$k(k-1)\binom{n}{k} = n \cdot (n-1)\binom{n-2}{k-2}$$

Proof (algebraic).

$$k(k-1)\frac{n!}{k!\,(n-k)!} = \frac{n!}{(k-2)!\,(n-k)!} = n(n-1) \cdot \frac{(n-2)!}{(k-2)!\,\big((n-2)-(k-2)\big)!} = n(n-1)\binom{n-2}{k-2}$$

- Sum of the first n natural numbers: For integers $n \geq 1$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

Proof (telescoping). For every k, $(k+1)^2 - k^2 = 2k+1$. Summing from $k=1$ to n, the left side telescopes:

$$(n+1)^2 - 1 = 2\sum_{k=1}^{n} k + n$$

Hence

$$2\sum_{k=1}^{n} k = n^2 + 2n - n = n(n+1)$$

which gives the result.
Proof (AP formula). The terms $1, 2, \ldots, n$ form an AP with $a_1 = 1$, $d = 1$, so by (2), $S_n = \frac{n}{2}[2 + (n-1)] = \frac{n(n+1)}{2}$.

- Sum of the first n squares: For integers $n \geq 1$,

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof (telescoping). For every k, $(k+1)^3 - k^3 = 3k^2 + 3k + 1$. Summing from $k=1$ to n, the left side telescopes:

$$(n+1)^3 - 1 = 3\sum_{k=1}^{n} k^2 + 3\sum_{k=1}^{n} k + n$$

Using $\sum k = \frac{n(n+1)}{2}$ and $(n+1)^3 - 1 - n = n(n+1)(n+2)$,

$$3\sum_{k=1}^{n} k^2 = n(n+1)(n+2) - \frac{3n(n+1)}{2} = \frac{n(n+1)(2n+1)}{2}$$

Dividing by 3 gives the result.
Proof (induction). For $n = 1$, both sides equal 1. Assume the formula holds for n. Then

$$\sum_{k=1}^{n+1} k^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)[n(2n+1) + 6(n+1)]}{6} = \frac{(n+1)(n+2)(2n+3)}{6}$$

which is the formula with $n+1$ in place of n, since $2n^2 + 7n + 6 = (n+2)(2n+3)$.

1 Arithmetic and Geometric Progressions

An arithmetic progression is defined by its starting term 'a_1' and unique common difference 'd'. A general AP can be written as a_1, a_2,..., a_n. Some useful results follow.

- Notice that the general AP can also be written as a_1, $a_1 + d$, $a_1 + 2d$,..., $a_1 + (n-1)d$.

$$a_n = a_1 + (n-1)d \tag{1}$$

- To sum the general AP, we can notice that the sum of the first and last terms is equal to the second and second-last terms and so on which is equal to $2a_1 + (n-1)d$ and we have $\frac{n}{2}$ pairs in the general AP.

$$S_n = \frac{n}{2}[2a_1 + (n-1)d] = \frac{n}{2}[a_1 + a_n] \tag{2}$$

A geometric progression is defined by its starting term 'u_1' and unique common ratio 'r'. A general GP can be written as u_1, u_2,..., u_n. Some useful results follow.

- Just like above, the general GP can be wriiten in a different way.

$$u_n = u_1 \cdot r^{n-1} \tag{3}$$

- To sum the general GP, we can shift the GP to the right and make its starting term u_2 by multiplying the entire GP by r. If we substract both these GPs we get a telescoping series.

$$S_n = u\left(\frac{1-r^n}{1-r}\right) = \frac{u_1 - u_n r}{1-r} \tag{4}$$

- Notice that if $|r| < 1$, the infinite sum seems to sum to a finite number. We can argue that as the value of n increases, the value of r^n decreases i.e. the limit r^n approaches as n goes to infinity is zero. None of the other terms in (4) care what the value of n is.

$$S_\infty = \frac{u}{1-r} \tag{5}$$

1. We can say three terms form an AP if $a_3 - a_2 = a_2 - a_1$. It comes out to be equal to the following term.

$$\frac{2\sqrt{b} - \sqrt{a} - \sqrt{c}}{2d}$$

Hence the new terms form an AP.

2. Notice that the sum can be written in an another way which gives us d in the denominator. This is a telescoping series where after simplification we end up with the following.

$$\frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$$

3. Same trick here. If we multiply with d, we get a telescoping series which after simplification gives us the following.

$$\frac{n-1}{a_1 a_n}$$

4. Follows from above. This telescoping series trick is only possible with the proposition that $a_1, a_2, ..., a_n$ are in an AP.

5. Let $P(n)$ denote the sum given.

$$P(n) = \sum_{i=0}^{n} (-1)^i C_n^i a_{i+1} = 0$$

We know that $a_{i+1} = a_1 + id$. So the sum can be written as follows.

$$\sum_{i=0}^{n} (-1)^i C_n^i [a_1 + id] = \sum_{i=0}^{n} (-1)^i C_n^i a_1 + \sum_{i=0}^{n} (-1)^i C_n^i id$$

From summation properties and the absorption identity ($k \cdot C_n^k = n \cdot C_{n-1}^{k-1}$), we can simplify as follows.

$$a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i + nd \cdot \sum_{i=1}^{n} (-1)^i C_{n-1}^{i-1}$$

By change of index from i to $k = i - 1$ in the second sum we can write the total sum as follows.

$$a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i + nd \cdot \sum_{k=0}^{n-1} (-1)^{k+1} C_{n-1}^k = a_1 \cdot \sum_{i=0}^{n} (-1)^i C_n^i - nd \cdot \sum_{k=0}^{n-1} (-1)^k C_{n-1}^k$$

By invoking the binomial theorem, we get the following result

$$a_1 \cdot 0 - nd \cdot 0 = 0 = P(n)$$

Hence, the proposition $P(n)$ is true for all $n \geq 2$ as $LHS = RHS$. Hence for any AP we have the equalities given by $P(n)$.

6. Same trick as above using the following property $k(k-1)C_n^k = n(n-1)C_{n-2}^{k-2}$.

7. Solve this using the given equality $\frac{1}{logk} + \frac{1}{logn} = \frac{2}{logm}$ as well as logarithmic properties. Take the log of both sides and transfer $logn$ to the right and then show LHS = RHS.

8. You have to show that the ratio $\frac{S_n}{S_{kn} - S_n}$ is independent of n and you will find it to be equal to $\frac{1}{k^2-1}$.

9. Starting with the first equation, using some manipulation of terms we can show,

$$x_1 = \frac{a}{n} - \left(\frac{n-1}{2}\right) d$$

Then in the second equation using $a_n = a_1 + kd$, we find the following (assuming $k = n - 1$),

$$\sum (x + kd)^2 = x^2 \sum 1 + d^2 \sum k^2 + 2xd \sum k$$

Simplifying this using $\sum a^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum a = \frac{n(n+1)}{2}$ and banishing k from above,

$$nx^2 + d^2 \left(\frac{n(n-1)(2n-1)}{6}\right) + xdn(n-1)$$

Substituting the value of x_1 we found earlier and simplifying gives us the following,

$$d = \pm \frac{2}{n} \sqrt{\frac{3(nb^2 - a^2)}{n^2 - 1}}$$

10. The number sequence corresponds to starting term 3 and common difference 3 so the nth term is $3n$ and sum of first n terms is $\frac{3n(n+1)}{2}$.

11. We know that the square of an odd natural number is given by $(2n+1)^2$ where n is any whole number. Assume an AP 1,2,3,...,n where $a = 1, d = 1$. Then each row is given by the following relation,

$$S_{3n+1} - S_n$$

Simplifying this with the sum of general AP formula we get

$$S_{3n+1} - S_n = (2n+1)^2$$

12. Notice if we divide a_{m+n} by a_{m-n} we get,

$$r^{2n} = \frac{A}{B} \implies r = \left(\frac{A}{B}\right)^{\frac{1}{2n}}$$

Also notice that if we multiply a_{m+n} and a_{m-n} we get,

$$a_m = \sqrt{AB}$$

Let $a = \frac{A}{r^{m+n-1}}$ then we can write a_n as

$$a_n = \frac{A}{r^m} = A\left(\frac{B}{A}\right)^{\frac{m}{2n}}$$

13. Notice that the sum $S_{2n} - S_n = a_{n+1} + a_{n+2} + ... + a_{2n} = r^n(a_1 + a_2 + ... + a_n) = r^n S_n$ and similarly $S_{3n} - S_{2n} = r^{2n} Sn$. Hence

$$\frac{S_n}{S_{2n} - S_n} = \frac{S_{2n} - S_n}{S_{3n} - S_{2n}} = \frac{1}{r^n}$$

14. Using the sum formula of a general GP we find that,

$$S_n = \frac{u(1-r^n)}{1-r}, \bar{S}_n = \frac{r^{1-n}}{a}\left(\frac{r^n - 1}{r-1}\right) \implies \frac{S_n}{\bar{S}_n} = a^2 r^{n-1}$$

Using $\sum_0^n i = \frac{n(n+1)}{2}$ and rules of exponents we can conclude that,

$$\prod = a^n r^{\frac{n(n-1)}{2}}$$

By observation we can conclude that

$$\left(\prod\right)^2 = \left(\frac{S_n}{\bar{S}_n}\right)^n$$

15. Let S be the required sum. Multiply S with x. We have a partial telescoping series which gives us

$$S - xS = (1 + x + x^2 + ... + x^n) - (n+1)x^{n+2}$$

After simplification and solving for S we get

$$S = \frac{1 - (n+2)x^{n+1} - (n+1)x^{n+2}}{(1-x)^2}$$

16. Let a_i represent a term in the sequence 1, 11, 111, ..., 111...1. Then we observe that: $\sum a_1 = 1$, $\sum a_2 = 12$, $\sum a_3 = 123$ and so on. Thus

$$\sum a_n = 123...n$$

17. Notice that we can write the series as:

$$nx + (n-1)x^2 + (n-2)x^3 + ... + 3x^{n-2} + 2x^{n-1} + 1x^n$$

$$(x^n + x^{n-1} + ... + x^2 + x) + (x^{n-1} + x^{n-2} + ... + x) + ... + (x^2 + x) + (x)$$

Notice that the sum of the given series is the sum of the same GP of terms n, n-1, ... 2, 1. Using the sum of GP formula we get:

$$\sum_1^n x\left(\frac{x^i - 1}{x-1}\right) = \frac{x}{x-1}\left(\sum_1^n x^i + \sum_1^n 1\right) = \frac{x^2(x^n - 1) - nx(x-1)}{(x-1)^2}$$

18. Let us denote the required sum as follows:

$$\mathbb{S} = \sum_{1}^{n} \frac{2i-1}{2^i} = \sum_{1}^{n} \frac{2i}{2^i} - \sum_{1}^{n} \frac{1}{2^i}$$

Using appropriate techniques we get the required sum:

$$\mathbb{S} = 3 - \frac{2n+3}{2^n}$$

19. Let a number in the sequence 49, 4489, 444889, ... be denoted by N_k. We claim that N_k is a perfect square for all k=0,1,2,... i.e. it is of the form $(number)^2$.

$$N_k = \underbrace{4...4}_{k}\underbrace{8...8}_{k-1}9 = \frac{4(10^k-1)}{9} \cdot 10^k + \frac{8(10^{k-1}-1)}{9} \cdot 10 + 9$$

If we take the common denominator over 9 and simplify we get the following for all k and N_k.

$$N_k = \left(\frac{2 \cdot 10^k + 1}{3}\right)^2$$

We can show that our base is a whole number. $10^k \equiv 1 \pmod 3 \implies 2 \cdot 10^k + 1 \equiv 0 \pmod 3$ meaning that $2 \cdot 10^k + 1$ is divisible by 3 and hence a whole number.

20. Let us denote any term in the geometric progression as a_i and the sum of subsequent terms as S.

$$a_i = q^{i-1} \quad \text{and} \quad S = S_\infty - S_i = \frac{q^i}{1-q}$$

We claim that $a_i = k \cdot S$.

$$k = \frac{a_i}{S} = q^{i-1} \cdot \frac{1-q}{q \cdot q^{i-1}} = \frac{1-q}{q}$$

This problem is solvable for $k \in (0, \infty) \because q < 1$.

21. Assume all x_i are nonzero real numbers, and that $(x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$ holds for every $n \geq 3$. To prove $x_1, x_2, x_3, ..., x_n, ...$ forms an infinite geometric progression by mathematical induction, we need to first review the base case n=3.

$$LHS = (x_1^2 + x_2^2)(x_2^2 + x_3^2) = x_1^2x_2^2 + x_2^2x_3^2 + x_1^2x_3^2 + x_2^4$$

$$RHS = (x_1x_2 + x_2x_3)^2 = x_1^2x_2^2 + x_2^2x_3^2 + 2x_1x_2^2x_3$$

$$LHS = RHS \implies x_1^2x_3^2 - 2x_1x_2^2x_3 + x_2^4 = (x_1x_3 - x_2^2)^2 = 0$$

From the above result, we can conclude $x_1x_3 - x_2^2 = 0 \implies \frac{x_2}{x_1} = \frac{x_3}{x_2} = r$ where we define r as the common ratio of a GP. Hence the base case of n=3 gives us a geometric progression.

Now we assume that the statement $(x_1, x_2, x_3, ..., x_n, ...$ forms an infinite geometric progression) holds for all terms upto x_k where $(k \geq 3)$, that is $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_k}{x_{k-1}} = r$. We need to prove it holds for $n = k+1$.

$$LHS = (x_1^2 + x_2^2 + ... + x_k^2)(x_2^2 + x_3^2 + ... + x_{k+1}^2)$$

$$RHS = (x_1x_2 + x_2x_3 + ... + x_kx_{k+1})^2$$

Let the first expression in parenthesis be written as $\sum a^2$ and the second in parenthesis be written as $\sum b^2$ in LHS and the expression in RHS as $(\sum ab)^2$. By Cauchy-Schwarz Inequality, $\left(\sum a^2\right)\left(\sum b^2\right) = \left(\sum ab\right)^2 \iff \frac{b_1}{a_1} = \frac{b_2}{a_2} = ...$ and substituting $a = x_i$ and $b = x_{i+1}$ we have $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_{k+1}}{x_k}$. Since $\frac{x_2}{x_1} = r$ by the inductive hypothesis, this common value is r, so $\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = \frac{x_{k+1}}{x_k} = r$. Hence the case of $n = k+1$ gives us a geometric progression. By mathematical induction:

$$\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = r \implies (x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$$

For the converse, if $x_{i+1} = rx_i$ for all i, then $b_i = ra_i$, so $\sum b^2 = r^2 \sum a^2$ and $\sum ab = r \sum a^2$, giving $\left(\sum a^2\right)\left(\sum b^2\right) = r^2\left(\sum a^2\right)^2 = \left(\sum ab\right)^2$. Hence the two statements are equivalent:

$$\frac{x_2}{x_1} = \frac{x_3}{x_2} = ... = r \iff (x_1^2 + x_2^2 + ... + x_{n-1}^2)(x_2^2 + x_3^2 + ... + x_n^2) = (x_1x_2 + x_2x_3 + ... + x_{n-1}x_n)^2$$

22. Assume a general AP and GP with starting terms a and b respectively and the common difference and common ratio as d and r respectively. The given conditions are $a_1 = b_1;\ a_2 = b_2;\ a_1 \neq a_2;\ a_n > 0 \implies d > 0;\ r > 1$. From $a_1 = b_1$ we get $a = b$, and from $a_2 = b_2$ we get $a + d = ar$. We can conclude that:

$$a = b = \frac{d}{r-1}$$

Let us evaluate the given expression. To prove $a_n < b_n$ for $n \geq 3$, it suffices to show $a_n - b_n < 0$.

$$a_n - b_n = a + (n-1)d - br^{n-1} = \frac{d}{r-1} \cdot \left(1 + (r-1)(n-1) - r^{n-1}\right)$$

Here $\frac{d}{r-1} > 0$, so we must show $1 + (r-1)(n-1) - r^{n-1} < 0 \iff 1 + (r-1)(n-1) < r^{n-1}$, which is true by Bernoulli's inequality, $(1+x)^m > 1 + mx$ for $x > -1,\ x \neq 0$ and integer $m \geq 2$, with $x = r - 1 > 0$ and $m = n - 1 \geq 2$. Hence:

$$a_n - b_n < 0 \iff a_n < b_n \quad \text{for all } n \geq 3 \quad (\text{for } n = 1, 2 \text{ equality holds, so } a_n = b_n.)$$

23. From the given conditions: $a_1 > 0$ and $\frac{a_2}{a_1} > 0 \implies r > 0$ and $b_2 - b_1 > 0 \implies d > 0$. Using logarithmic properties and proper grouping we can write:

$$log_\phi a_n - b_n = log_\phi(a_1 \cdot q^{n-1}) - (b + (n-1)d) = (log_\phi a_1 - b_1) + ((n-1) \cdot (log_\phi q - d))$$

For the expression to be independent of n, the terms containing n must be zero. Hence:

$$log_\phi q - d = 0 \iff \phi = q^{\frac{1}{d}}$$